

```
!pip install --upgrade datasets
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.29.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=21.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (24.0.0)
Requirement already satisfied: dill<0.4.2,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: httpx<1.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.28.1)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.7.0)
Requirement already satisfied: multiprocessing<0.70.20 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2026.4.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2026.4.0) (2026.4.0)
Requirement already satisfied: huggingface-hub<2.0,>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.18.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!<4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2026.4.0) (4.0.0a1)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (2026.5.20)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (3.18)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0.0->datasets) (0.14.0)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (8.1.8)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (1.0.15)
Requirement already satisfied: typer<0.26.0,>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (0.26.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.25.0->datasets) (4.12.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.31.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2026.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (1.5.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (6.1.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!<4.0.0a1->fsspec) (1.18.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0->huggingface-hub) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0->huggingface-hub) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer<0.26.0,>=0.20.0->huggingface-hub) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<0.26.0->huggingface-hub) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer<0.26.0->huggingface-hub) (2.18.1)
Requirement already satisfied: mdurl~>0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->huggingface-hub) (0.1.2)
```

Task 1: Load & Explore Dataset

```
from datasets import load_dataset
```

```
try:
```

```
    dataset = load_dataset("ag_news")
```

```
except:
```

```
    dataset = load_dataset("SetFit/ag_news")
```

```
print(dataset)
```

```
train.jsonl: 100%
```

```
33.8M/33.8M [00:01<00:00, 27.4MB/s]
```

```
test.jsonl: 100%
```

```
2.13M/2.13M [00:00<00:00, 6.09MB/s]
```

```
Generating train split: 100%
```

```
120000/120000 [00:00<00:00, 410293.72 examples/s]
```

```
Generating test split: 100%
```

```
7600/7600 [00:00<00:00, 183765.57 examples/s]
```

```
DatasetDict({
  train: Dataset({
    features: ['text', 'label', 'label_text'],
    num_rows: 120000
  })
  test: Dataset({
    features: ['text', 'label', 'label_text'],
    num_rows: 7600
  })
})
```

```
!pip install tensorflow matplotlib scikit-learn
```

```

Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
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Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (0.2.0)
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Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.2.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (1.81.0)
Requirement already satisfied: tensorboard~2.20.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow) (3.13.2)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow) (0.43.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow) (0.19.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow) (2025.11.12)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (3.7.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (0.7.0)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from tensorboard~2.20.0->tensorflow) (3.1.0)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from werkzeug>=1.0.1->tensorboard~2.20.0->tensorflow) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.10.0->tensorflow) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.10.0->tensorflow) (2.19.2)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)

```

Task 2 - Text preprocessing

```

# Imports
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from datasets import load_dataset
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, SimpleRNN, LSTM, GRU, Dense, Bidirectional
from tensorflow.keras.optimizers import Adam
from sklearn.model_selection import train_test_split

```

```

# 2. Number of samples
print("Training samples:", len(dataset['train']))
print("Test samples:", len(dataset['test']))

```

```

# Convert to pandas for easy manipulation
train_df = pd.DataFrame(dataset['train'])
test_df = pd.DataFrame(dataset['test'])

```

```

# 3. Show 5 samples
print(train_df[['text', 'label']].head())

```

```

# 4. Class distribution

```

```

label_names = ["World", "Sports", "Business", "Sci/Tech"]
train_df['label_name'] = train_df['label'].apply(lambda x: label_names[x])

dist = train_df['label_name'].value_counts()
print(dist)

# 5. Plot
plt.figure()
dist.plot(kind='bar')
plt.title("Class Distribution")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()

```

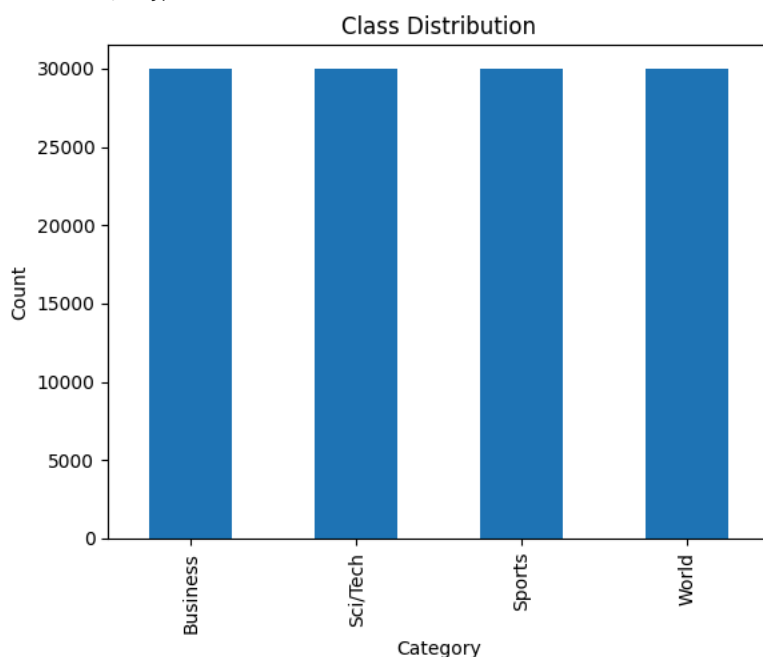
Training samples: 120000

Test samples: 7600

```

      text  label
0  Wall St. Bears Claw Back Into the Black (Reute...    2
1  Carlyle Looks Toward Commercial Aerospace (Reu...    2
2  Oil and Economy Cloud Stocks' Outlook (Reuters...    2
3  Iraq Halts Oil Exports from Main Southern Pipe...    2
4  Oil prices soar to all-time record, posing new...    2
label_name
Business    30000
Sci/Tech    30000
Sports      30000
World       30000
Name: count, dtype: int64

```



Task 3 : Tokenization and padding

```

# Extract text and labels
X_train = train_df['text']
y_train = train_df['label']
X_test = test_df['text']
y_test = test_df['label']

# Tokenization
vocab_size = 20000
tokenizer = Tokenizer(num_words=vocab_size, oov_token="<OOV>")
tokenizer.fit_on_texts(X_train)

# Convert to sequences
X_train_seq = tokenizer.texts_to_sequences(X_train)
X_test_seq = tokenizer.texts_to_sequences(X_test)

# Padding
max_len = 100
X_train_pad = pad_sequences(X_train_seq, maxlen=max_len, padding='post')
X_test_pad = pad_sequences(X_test_seq, maxlen=max_len, padding='post')

```

```
# Labels (no need for one-hot since using sparse_categorical_crossentropy)
print("Train shape:", X_train_pad.shape)
print("Test shape:", X_test_pad.shape)
print("Train labels:", y_train.shape)
print("Test labels:", y_test.shape)
```

```
Train shape: (120000, 100)
Test shape: (7600, 100)
Train labels: (120000,)
Test labels: (7600,)
```

Task 4 - Simple RNN Model

```
model_rnn = Sequential([
    Embedding(vocab_size, 128, input_length=max_len),
    SimpleRNN(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_rnn.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

history_rnn = model_rnn.fit(X_train_pad, y_train,
                           epochs=5,
                           batch_size=128,
                           validation_split=0.2)

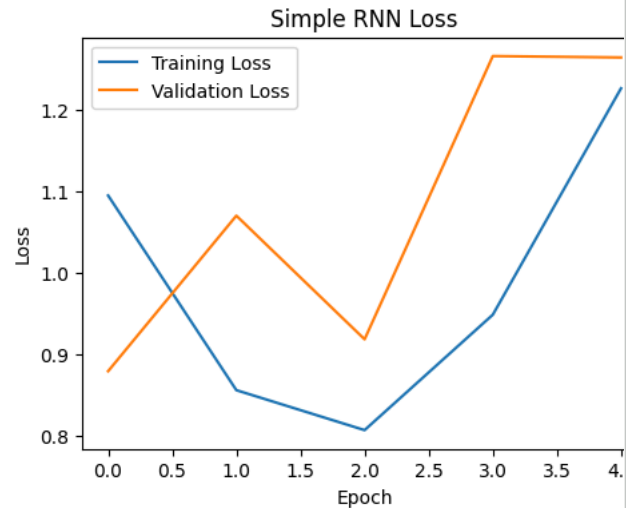
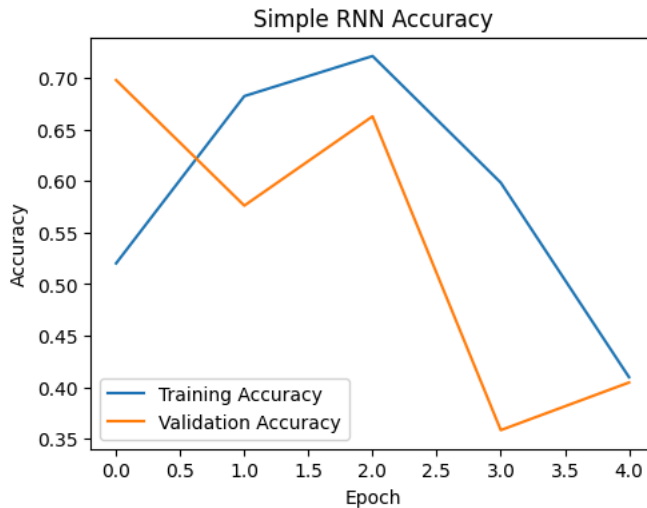
# Evaluation
test_loss, test_acc = model_rnn.evaluate(X_test_pad, y_test)
print("Test Accuracy:", test_acc)

def plot_history(history, model_name):
    plt.figure(figsize=(12, 4))
    plt.subplot(1, 2, 1)
    plt.plot(history.history['accuracy'], label='Training Accuracy')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
    plt.title(f'{model_name} Accuracy')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.subplot(1, 2, 2)
    plt.plot(history.history['loss'], label='Training Loss')
    plt.plot(history.history['val_loss'], label='Validation Loss')
    plt.title(f'{model_name} Loss')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.show()

plot_history(history_rnn, "Simple RNN")
```

```
Epoch 1/5
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is deprecated
warnings.warn(
750/750 ————— 58s 73ms/step - accuracy: 0.5202 - loss: 1.0943 - val_accuracy: 0.6978 - val_loss: 0.8787
Epoch 2/5
750/750 ————— 55s 73ms/step - accuracy: 0.6823 - loss: 0.8554 - val_accuracy: 0.5762 - val_loss: 1.0695
Epoch 3/5
750/750 ————— 56s 74ms/step - accuracy: 0.7212 - loss: 0.8063 - val_accuracy: 0.6627 - val_loss: 0.9178
Epoch 4/5
750/750 ————— 55s 73ms/step - accuracy: 0.5984 - loss: 0.9480 - val_accuracy: 0.3585 - val_loss: 1.2655
Epoch 5/5
750/750 ————— 84s 76ms/step - accuracy: 0.4099 - loss: 1.2259 - val_accuracy: 0.4047 - val_loss: 1.2637
238/238 ————— 2s 7ms/step - accuracy: 0.4072 - loss: 1.2521
Test Accuracy: 0.40723684430122375
```



Task 5 - LSTM Model

```
model_lstm = Sequential([
    Embedding(vocab_size, 128),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

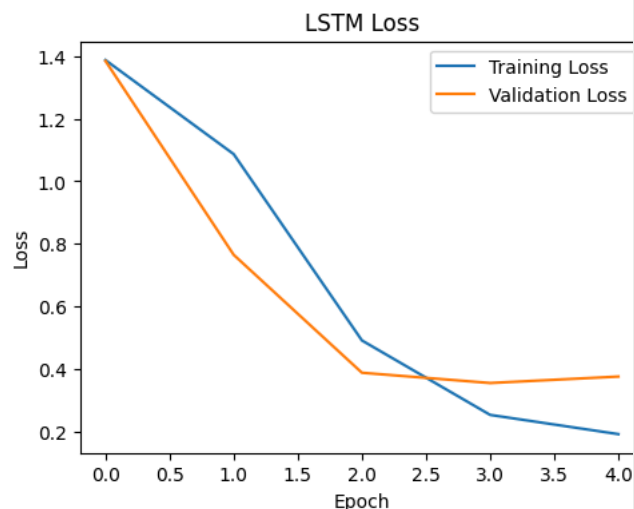
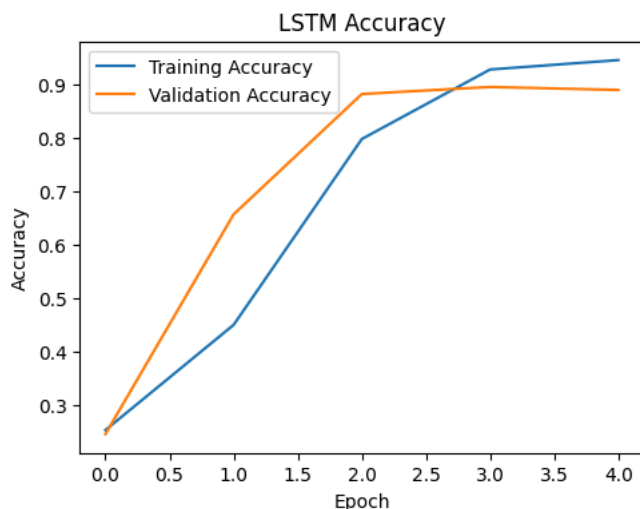
model_lstm.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

history_lstm = model_lstm.fit(X_train_pad, y_train,
                             epochs=5,
                             batch_size=128,
                             validation_split=0.2)

test_loss, test_acc = model_lstm.evaluate(X_test_pad, y_test)
print("LSTM Test Accuracy:", test_acc)

plot_history(history_lstm, "LSTM")
```

Epoch 1/5
 750/750 — 159s 209ms/step - accuracy: 0.2522 - loss: 1.3878 - val_accuracy: 0.2447 - val_loss: 1.3850
 Epoch 2/5
 750/750 — 190s 192ms/step - accuracy: 0.4496 - loss: 1.0870 - val_accuracy: 0.6568 - val_loss: 0.7649
 Epoch 3/5
 750/750 — 202s 192ms/step - accuracy: 0.7981 - loss: 0.4910 - val_accuracy: 0.8824 - val_loss: 0.3877
 Epoch 4/5
 750/750 — 143s 191ms/step - accuracy: 0.9283 - loss: 0.2529 - val_accuracy: 0.8955 - val_loss: 0.3548
 Epoch 5/5
 750/750 — 144s 192ms/step - accuracy: 0.9460 - loss: 0.1914 - val_accuracy: 0.8900 - val_loss: 0.3755
 238/238 — 4s 17ms/step - accuracy: 0.9066 - loss: 0.3174
 LSTM Test Accuracy: 0.9065789580345154



Task 6 - GRU Model

```
model_gru = Sequential([
    Embedding(vocab_size, 128),
    GRU(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

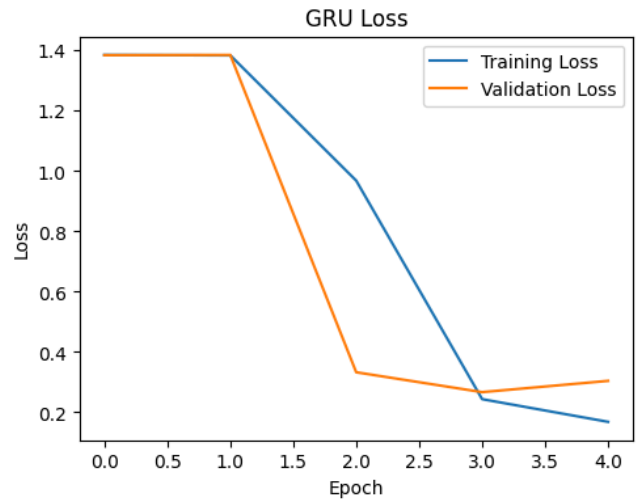
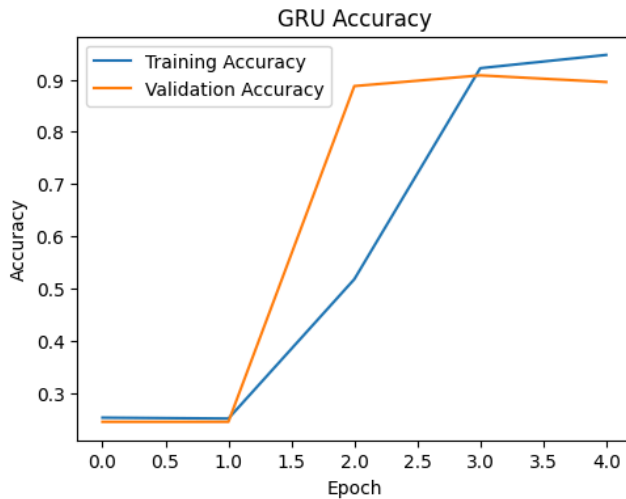
model_gru.compile(loss='sparse_categorical_crossentropy',
                  optimizer='adam',
                  metrics=['accuracy'])

history_gru = model_gru.fit(X_train_pad, y_train,
                            epochs=5,
                            batch_size=128,
                            validation_split=0.2)

test_loss, test_acc = model_gru.evaluate(X_test_pad, y_test)
print("GRU Test Accuracy:", test_acc)

plot_history(history_gru, "GRU")
```

Epoch 1/5
 750/750 — 161s 211ms/step - accuracy: 0.2529 - loss: 1.3855 - val_accuracy: 0.2449 - val_loss: 1.3846
 Epoch 2/5
 750/750 — 153s 204ms/step - accuracy: 0.2512 - loss: 1.3840 - val_accuracy: 0.2449 - val_loss: 1.3854
 Epoch 3/5
 750/750 — 153s 203ms/step - accuracy: 0.5179 - loss: 0.9684 - val_accuracy: 0.8875 - val_loss: 0.3327
 Epoch 4/5
 750/750 — 153s 205ms/step - accuracy: 0.9217 - loss: 0.2435 - val_accuracy: 0.9080 - val_loss: 0.2668
 Epoch 5/5
 750/750 — 154s 206ms/step - accuracy: 0.9471 - loss: 0.1683 - val_accuracy: 0.8953 - val_loss: 0.3042
 238/238 — 3s 14ms/step - accuracy: 0.9070 - loss: 0.2826
 GRU Test Accuracy: 0.90697365992218



Task 7 - Bi-LSTM Model

```
model_bilstm = Sequential([
    Embedding(vocab_size, 128),
    Bidirectional(LSTM(64)),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_bilstm.compile(loss='sparse_categorical_crossentropy',
                    optimizer='adam',
                    metrics=['accuracy'])

history_bilstm = model_bilstm.fit(X_train_pad, y_train,
                                  epochs=5,
                                  batch_size=128,
                                  validation_split=0.2)

test_loss, test_acc = model_bilstm.evaluate(X_test_pad, y_test)
print("BiLSTM Test Accuracy:", test_acc)

plot_history(history_bilstm, "BiLSTM")
```

```

Epoch 1/5
750/750 ————— 260s 342ms/step - accuracy: 0.8724 - loss: 0.3633 - val_accuracy: 0.9084 - val_loss: 0.2693
Epoch 2/5
750/750 ————— 249s 332ms/step - accuracy: 0.9369 - loss: 0.1903 - val_accuracy: 0.9057 - val_loss: 0.2744
Epoch 3/5
750/750 ————— 248s 330ms/step - accuracy: 0.9509 - loss: 0.1412 - val_accuracy: 0.9015 - val_loss: 0.3053
Epoch 4/5
750/750 ————— 249s 332ms/step - accuracy: 0.9623 - loss: 0.1080 - val_accuracy: 0.8952 - val_loss: 0.3331
Epoch 5/5
750/750 ————— 251s 334ms/step - accuracy: 0.9704 - loss: 0.0807 - val_accuracy: 0.8922 - val_loss: 0.3797
238/238 ————— 6s 24ms/step - accuracy: 0.9059 - loss: 0.3373
BiLSTM Test Accuracy: 0.9059210419654846

```



```

# Without clipping
opt1 = Adam()
model_no_clip = model_lstm

# With clipping
opt2 = Adam(clipnorm=1.0)

model_clip = Sequential([
    Embedding(vocab_size, 128),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(4, activation='softmax')
])

model_clip.compile(loss='sparse_categorical_crossentropy',
                  optimizer=opt2,
                  metrics=['accuracy'])

history_clip = model_clip.fit(X_train_pad, y_train,
                             epochs=5,
                             batch_size=128,
                             validation_split=0.2)

plot_history(history_clip, "LSTM with Gradient Clipping")

```

```

Epoch 1/5
750/750 ————— 150s 197ms/step - accuracy: 0.2531 - loss: 1.3857 - val_accuracy: 0.2617 - val_loss: 1.3844
Epoch 2/5
750/750 ————— 145s 193ms/step - accuracy: 0.2532 - loss: 1.3843 - val_accuracy: 0.2351 - val_loss: 1.3846
Epoch 3/5
750/750 ————— 146s 194ms/step - accuracy: 0.6710 - loss: 0.7267 - val_accuracy: 0.8953 - val_loss: 0.3257
Epoch 4/5
750/750 ————— 154s 206ms/step - accuracy: 0.9270 - loss: 0.2431 - val_accuracy: 0.9010 - val_loss: 0.3015
Epoch 5/5
750/750 ————— 193s 193ms/step - accuracy: 0.9471 - loss: 0.1737 - val_accuracy: 0.8992 - val_loss: 0.3047

```

